

Volatility Targeting in the Taiwan Stock Market: Leverage Amplification, Model Selection, and Practical Implementation*

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Abstract

We examine volatility targeting (VT) strategies in the Taiwan stock market using the FTSE TWSE Taiwan 50 Index ETF (0050.TW) over 2008–2026. Three contributions emerge. First, the broad TAIEX exhibits a substantially stronger diversification amplification of the leverage effect ($4.6\times$ for ~ 900 stocks; $1.45\times$ for the investable 0050.TW, 50 stocks) compared to the U.S. market ($2.8\times$), attributable to higher correlation asymmetry among Taiwan-listed stocks during market declines. Despite the strong index-level leverage effect ($\gamma = 0.272$, $t = 3.18$), GJR-GARCH does not significantly improve QLIKE over standard GARCH (DM $p = 0.86$), yet GJR-based VT produces meaningfully higher Sharpe ratios ($+0.114$)—suggesting that tail-specific forecast accuracy, rather than average accuracy, drives VT performance. Second, a simple EWMA-based VT strategy improves risk-adjusted returns (Sharpe $0.729 \rightarrow 0.796$) while halving maximum drawdown ($-41.3\% \rightarrow -18.4\%$), with Student- t VaR achieving a 0.5% violation rate (Basel III Green Zone). Third, we discover a time-zone arbitrage strategy exploiting the information gap between U.S. and Asian market closes: a 10-day SPY momentum signal applied to 0050.TW

*Data and replication code available upon request.

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generates a net Sharpe ratio of 1.47 ($t = 3.76$, exceeding the $t > 3.0$ threshold of Harvey et al., 2016), validated across six Asia-Pacific markets, five out-of-sample sub-periods, and Bonferroni-corrected tests. Additionally, we show that the month-over-month change in Taiwan’s leading indicator composite is the strongest local macro predictor of realized volatility ($t = 3.74$, $R^2 = 7.1\%$), and that combining VIX scaling with leading indicator direction significantly improves the VIX-only strategy (full-sample Sharpe $0.999 \rightarrow 1.151$, DM $p = 0.0005$), though this monthly-frequency signal cannot enhance the daily-frequency momentum strategy. The pure 8.63/VIX strategy achieves a corrected Sharpe ratio of 0.690 over 2016–2026 with monthly rebalancing and lagged weights. These results demonstrate that VT benefits extend to emerging Asian markets and that cross-market information dynamics create persistent trading opportunities in time-zone-separated markets.

Keywords: volatility targeting, Taiwan stock market, leverage effect, diversification amplification, GARCH, time-zone arbitrage, leading indicators, VaR

JEL Classification: C22, C53, G11, G15

1 Introduction

Volatility targeting (VT), the practice of dynamically scaling portfolio exposure inversely to predicted volatility, has attracted considerable attention following the influential work of [Moreira and Muir \[2017\]](#). The central insight is intuitive: investors should hold less when markets are turbulent and more when markets are calm. Studies have documented that this approach reduces maximum drawdown substantially, though its impact on risk-adjusted returns remains debated [[Fleming et al., 2001](#), [Harvey et al., 2018](#)].

Despite a growing literature, virtually all evidence on VT strategies pertains to U.S. markets or, more broadly, to developed Western economies. The applicability of these findings to Asian emerging markets—where microstructure, investor composition, and information transmission dynamics differ materially—remains an open question. This gap is particularly surprising in the case of Taiwan, which ranks among the world’s most active equity markets by turnover ratio and hosts a large retail investor base that could

benefit directly from practical VT implementations.

The Taiwan stock market presents several features that make it a compelling laboratory for VT research. First, the TAIEX exhibits a leverage effect—the asymmetric volatility response to positive and negative shocks formalized by Black [1976] and Glosten et al. [1993]—that is substantially stronger at the index level than at the individual stock level. We document that this *diversification amplification* reaches $4.6\times$ for the broad TAIEX (~ 900 stocks) relative to individual stocks, though a more modest $1.45\times$ for the investable 0050.TW ETF (50 stocks), compared to $2.8\times$ in the United States, a difference we attribute to higher correlation asymmetry among Taiwanese equities during market declines [Ang and Chen, 2002]. Second, the absence of a deep, liquid local implied volatility index (the VIXTWN was introduced only in November 2020 with limited history) forces practitioners to rely on either model-based volatility estimates or proxy measures such as the U.S. VIX. Third, Taiwan’s geographic position—trading hours that open after the U.S. market closes—creates a natural information transmission channel that we exploit in what we term a *time-zone arbitrage* strategy.

This paper makes three contributions.

First, we document the leverage effect and its diversification amplification in the Taiwan market, extending the analysis of Ang and Chen [2002] to an Asian emerging-market context. Using GJR-GARCH estimation on the TAIEX ($\gamma = 0.272$, $t = 3.18$), ten individual Taiwanese stocks, and the 0050.TW ETF ($\gamma = 0.087$), we show that the broad TAIEX gamma exceeds the average individual stock gamma by a factor of $4.6\times$, though the investable 0050.TW (50 stocks) shows a more modest amplification of $1.45\times$. This amplification arises because stock correlations increase asymmetrically during downturns, a phenomenon first identified by Ang and Chen [2002] for U.S. equities. We further establish a strong cross-market spillover channel: lagged SPY returns predict next-day 0050.TW returns with a correlation of $r = 0.376$, and VIX Granger-causes Taiwan equity volatility ($F = 58.8$, $p < 0.001$).

Second, we evaluate multiple VT strategies adapted for Taiwan, including EWMA-based VT, GJR-GARCH VT, and a VIX-proxy approach (8.63/VIX) calibrated using

the VIXTWN-to-VIX ratio of 1.39. The EWMA strategy improves the Sharpe ratio from 0.729 (buy-and-hold) to 0.796 while halving maximum drawdown from -41.3% to -18.4% . We show that Student- t Value-at-Risk achieves a 0.5% violation rate over 2020–2026, within the Basel III Green Zone (noting that the traffic-light framework is defined for 250-day windows; we apply the 1% threshold as a benchmark across our full sample). A key finding is that despite the statistically significant leverage effect, GJR-GARCH does not significantly improve QLIKE over standard GARCH (Diebold-Mariano $p = 0.86$), yet GJR-based VT produces a meaningfully higher Sharpe ratio (+0.114), suggesting that tail-specific forecast accuracy—rather than average forecast accuracy—drives VT performance.

Third, we document and test a time-zone arbitrage strategy based on U.S. equity momentum. The key observation is that Taiwan’s market opens approximately 12 hours after the U.S. close, creating an information gap that local prices do not fully incorporate at the open. A 10-day SPY momentum signal applied to 0050.TW generates a net Sharpe ratio of 1.47 with a Harvey t -statistic of 3.76, exceeding the $t > 3.0$ threshold recommended by [Harvey et al. \[2016\]](#) for controlling multiple testing. This effect is not an artifact of U.S.-listed Taiwan ETFs (the EWT control fails), confirming that the alpha originates from the time-zone information gap rather than from common factor exposure. We validate this finding across five out-of-sample sub-periods, six Asia-Pacific markets, and through bootstrap confidence intervals and Bonferroni-corrected tests.

The remainder of the paper proceeds as follows. Section 2 describes the data and methodology. Section 3 examines the leverage effect and diversification amplification. Section 4 evaluates VT strategies. Section 5 presents the time-zone arbitrage strategy. Section 6 explores macroeconomic indicators for Taiwan. Section 7 addresses VaR and risk management. Section 8 discusses practical implications. Section 9 concludes.

2 Data and Methodology

2.1 Data Sources

We use daily closing prices, adjusted for dividends and splits, for the following instruments obtained from Yahoo Finance and the Taiwan Stock Exchange (TWSE):

- **0050.TW**: Yuanta/P-shares Taiwan Top 50 ETF, tracking the FTSE TWSE Taiwan 50 Index. Sample: January 2008 to March 2026 (4,532 trading days).
- **TWII (TAIEX)**: Taiwan Capitalization Weighted Stock Index. Sample: January 1997 to March 2026 (7,148 trading days).
- **Ten individual Taiwan stocks**: Taiwan Semiconductor (2330.TW), Hon Hai Precision (2317.TW), MediaTek (2454.TW), Cathay Financial (2882.TW), Mega Financial (2886.TW), CTBC Financial (2891.TW), Chunghwa Telecom (2412.TW), Yuanta Financial (2885.TW), Fubon Financial (2881.TW), and Yuanta High Dividend ETF (0056.TW, an ETF tracking mid-cap dividend stocks).
- **U.S. benchmarks**: S&P 500 (SPY), CBOE Volatility Index (VIX), CBOE Emerging Markets ETF Volatility Index (VXEEM), and the iShares MSCI Taiwan ETF (EWT).
- **Asia-Pacific markets**: Nikkei 225 (Japan), Hang Seng (Hong Kong), ASX 200 (Australia), Straits Times (Singapore), KOSPI (South Korea), and Sensex (India).
- **VIXTWN**: TAIEX Options Volatility Index, compiled by the Taiwan Futures Exchange (TAIFEX) using the CBOE VIX methodology. Available from November 2020.

Log returns are computed as $r_t = \ln(P_t/P_{t-1}) \times 100$. Table 1 reports summary statistics.

Table 1: Summary Statistics for Key Assets

	Mean (%)	Std (%)	Skewness	Kurtosis	γ_{GJR}	$t(\gamma)$
TWII (1997–2026)	0.019	1.45	−0.31	5.82	0.272	3.18
0050.TW (2008–2026)	0.034	1.38	−0.47	4.73	0.087	2.20
SPY (2008–2026)	0.042	1.18	−0.52	12.30	0.211	5.79
TSMC (2330.TW)	0.051	1.92	−0.18	3.41	0.039	0.87
10-stock average	0.028	1.71	−0.22	3.68	0.060	—

Notes: γ_{GJR} is the asymmetric leverage parameter from GJR-GARCH(1,1) with $w = 2000$ rolling window. $t(\gamma)$ uses Newey-West HAC standard errors. 10-stock average is the equal-weighted mean of the ten individual Taiwanese stocks listed in Section 2.

2.2 Volatility Models

We employ three volatility models, estimated via rolling-window maximum likelihood.

GJR-GARCH(1,1) [Glosten et al., 1993]:

$$\sigma_t^2 = \omega + (\alpha + \gamma \cdot \mathbb{1}_{r_{t-1} < 0}) r_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (1)$$

where $\gamma > 0$ captures the leverage effect: negative returns increase conditional variance more than positive returns of equal magnitude.

GARCH(1,1) [Bollerslev, 1986]: the restriction $\gamma = 0$ of Equation (1).

EWMA (Exponentially Weighted Moving Average):

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2 \quad (2)$$

with decay factor $\lambda = 0.94$ (the RiskMetrics default). EWMA requires no parameter estimation and can be implemented without optimization software.

All GARCH and GJR-GARCH models are estimated using a rolling window of $w = 2000$ trading days, following Hwang and Valls Pereira [2006] who demonstrate that win-

dows below 500 induce substantial persistence bias. We verify that our core results are robust to $w = 504$ and $w = 1000$.

2.3 Forecast Evaluation

We evaluate volatility forecasts using the QLIKE loss function [Patton, 2011]:

$$\text{QLIKE}_t = \ln(\sigma_t^2) + \frac{r_t^2}{\sigma_t^2} \quad (3)$$

which is robust to noise in the volatility proxy r_t^2 and penalizes underprediction more heavily than overprediction. Model comparisons use the Diebold and Mariano [1995] test with Newey-West standard errors for the null hypothesis of equal predictive ability.

2.4 Volatility Targeting Strategies

A volatility targeting strategy sets the portfolio weight as:

$$w_t = \frac{\sigma_{\text{target}}}{\hat{\sigma}_t} \quad (4)$$

where σ_{target} is the annualized target volatility and $\hat{\sigma}_t$ is the volatility forecast. The weight is capped at 100% (no leverage). The portfolio return is:

$$r_t^{\text{VT}} = w_{t-1} \cdot r_t + (1 - w_{t-1}) \cdot r_t^f \quad (5)$$

where r_t^f is the risk-free return (proxied by the 3-month Treasury bill rate) and the weight uses information available at $t - 1$ only, avoiding look-ahead bias.

For the VIX-proxy strategy, the weight is:

$$w_t = \min\left(\frac{K}{\text{VIX}_t}, 1\right) \quad (6)$$

where $K = 8.63 = 12/1.39$ adjusts the standard 12/VIX formula [Bozovic, 2024] by the VIXTWN-to-VIX ratio of 1.39. The VIX is observed at the U.S. close, and the weight

is applied to the Taiwan trading session that begins the following calendar day, ensuring no contemporaneous overlap.

2.5 VIX as a Proxy for Taiwan Implied Volatility

A practical question for Taiwan VT is whether the U.S. VIX adequately captures Taiwan-specific volatility dynamics. Using the 64 months of overlapping data between VIX and VIXTWN (November 2020 to March 2026), we find a Spearman rank correlation of 0.595 between VIX and subsequent 0050.TW realized volatility, compared to 0.459 for VXEEM (the CBOE Emerging Markets Volatility Index). The difference is statistically significant (Steiger $Z = 16.2$, $p < 0.001$, based on dependent correlation comparison of daily-frequency observations within the 64-month overlap period). This result is initially counterintuitive—Taiwan is an emerging market, yet the developed-market VIX is a better predictor than the emerging-market VXEEM. The explanation lies in the composition of 0050.TW, which is dominated by Taiwan Semiconductor Manufacturing Company (TSMC), accounting for approximately 50% of the index weight. TSMC’s fortunes are closely tied to global semiconductor demand and U.S. technology sector sentiment, making U.S. VIX a natural proxy for the risk factors most relevant to the Taiwanese market.

The VIXTWN-to-VIX ratio averages 1.393 with a coefficient of variation of 10%, reflecting the higher baseline volatility of the Taiwanese market relative to the S&P 500. We calibrate $K = 12/1.39 = 8.63$ so that a VIX level of 20 (approximately the long-run median) produces a weight of $8.63/20 = 43\%$, a prudent allocation for an emerging-market equity position.

2.6 Transaction Costs and Rebalancing

For 0050.TW, we apply a round-trip transaction cost of 0.585%, reflecting the Taiwan Securities Transaction Tax (0.30% on sell) plus estimated brokerage commissions (0.1425% per side, discounted). Unless otherwise stated, all VT strategies use monthly rebalancing. We have verified that daily rebalancing improves the Sharpe ratio marginally but increases turnover and transaction costs substantially. Monthly rebalancing represents a

practical specification for individual investors.

3 The Leverage Effect in the Taiwan Market

3.1 Index-Level Leverage

The leverage effect—the tendency for volatility to rise more following negative returns than positive returns of equal magnitude—is a well-established stylized fact in developed equity markets [Black, 1976, Christie, 1982, Glosten et al., 1993]. We examine whether this phenomenon extends to Taiwan and, if so, whether its magnitude differs from the U.S. benchmark.

Table 2 reports GJR-GARCH γ estimates for the TAIEX, 0050.TW, SPY, and ten individual Taiwanese stocks. The TAIEX exhibits a statistically significant leverage effect ($\gamma = 0.272$, $t = 3.18$), substantially exceeding the U.S. S&P 500 ($\gamma = 0.211$, $t = 5.79$). The 0050.TW ETF, which tracks the 50 largest Taiwanese companies, shows a smaller but still significant $\gamma = 0.087$ ($t = 2.20$). The difference between the broad market index (TAIEX, 900+ stocks) and the concentrated ETF (50 stocks) is informative: the broader index captures more of the small- and mid-cap stocks whose correlations increase more dramatically during downturns.

Table 2: GJR-GARCH Leverage Parameters Across Taiwan and U.S. Markets

Asset	γ	t -stat	α	β	Persistence
TWII (TAIEX)	0.272	3.18	0.012	0.870	0.990
0050.TW	0.087	2.20	0.025	0.930	0.987
SPY	0.211	5.79	0.000	0.897	0.993
<i>Individual Taiwan Stocks</i>					
TSMC (2330)	0.039	0.87	0.031	0.944	0.988
Hon Hai (2317)	0.052	1.14	0.028	0.939	0.985
MediaTek (2454)	0.044	0.96	0.033	0.935	0.984
Mega Financial (2886)	0.179	2.42	0.015	0.901	0.977
0056 ETF	0.112	1.87	0.021	0.922	0.982
10-stock average	0.060	—	0.027	0.933	0.984

Notes: GJR-GARCH(1,1) with rolling window $w = 2000$. t -statistics use Newey-West HAC standard errors. Persistence = $\alpha + \beta + \gamma/2$. Individual stocks use the full available sample (2008–2026).

3.2 Diversification Amplification

A striking finding is that the average γ across the ten individual Taiwanese stocks is only 0.060, while the TAIEX index γ is 0.272—a ratio of $4.6\times$. This ratio should be interpreted cautiously given the small cross-section ($N = 10$) and the inclusion of one ETF (0056.TW). We term this the *diversification amplification* of the leverage effect: the process of aggregating individual stocks into an index magnifies the asymmetric volatility response. For comparison, the analogous ratio for U.S. equities is $2.8\times$ (SPY $\gamma = 0.211$ versus an average individual stock γ of 0.079, based on the 50 largest S&P 500 constituents).

An important caveat is that the $4.6\times$ amplification reflects the *broad* TAIEX (~ 900 stocks) relative to individual stocks. When using the investable 0050.TW ETF (50 stocks, $\gamma = 0.087$) as the index measure, the amplification ratio is a more modest

$0.087/0.060 = 1.45\times$. The difference between the TAIEX-based ($4.6\times$) and 0050-based ($1.45\times$) ratios reflects the broader TAIEX’s inclusion of small- and mid-cap stocks whose correlations exhibit stronger asymmetry during downturns. For practitioners implementing VT on the investable 0050.TW, the effective leverage amplification is therefore substantially lower than the headline $4.6\times$ figure, and GJR-GARCH’s advantage over symmetric GARCH is correspondingly attenuated (consistent with the insignificant DM test reported in Section 4).

The mechanism underlying this amplification is correlation asymmetry [Ang and Chen, 2002]: stock-level correlations increase during market declines and decrease during market advances. When the market falls, stocks move more in lockstep, amplifying portfolio-level variance beyond what individual stock variances alone would imply. The leverage effect at the index level therefore reflects not only individual stock asymmetries but also the asymmetric correlation structure of the cross-section.

To verify this mechanism, we compute the average pairwise correlation among the ten Taiwanese stocks separately for down days ($r_{\text{market}} < 0$) and up days ($r_{\text{market}} > 0$). The down-day correlation exceeds the up-day correlation by an average of 0.042 for U.S. stocks and 0.071 for Japanese stocks [Ang and Chen, 2002]. For our Taiwan sample, the difference is 0.058, intermediate between the U.S. and Japanese values and consistent with the Taiwan amplification ratio falling between these markets. While we do not have access to full intraday correlation data for all ten stocks, the observed amplification pattern is fully consistent with the correlation asymmetry hypothesis.

The broad TAIEX amplification ratio of $4.6\times$ exceeds that of the U.S. ($2.8\times$), which we attribute to two factors: (1) the Taiwan market is more concentrated (TSMC alone accounts for $\sim 50\%$ of 0050.TW), creating a “dominant stock” effect where TSMC’s price moves mechanically drive the index, and (2) the retail-investor-heavy structure of the Taiwanese market may generate stronger asymmetric correlation dynamics during market declines, consistent with the downside correlation literature [Ang and Chen, 2002]. However, for the investable 0050.TW ETF, the amplification is only $1.45\times$, suggesting that the broad TAIEX figure is driven substantially by small- and mid-cap stocks not

included in the top-50 index. This finding has direct implications for model selection: while broad-index VT strategies should account for the leverage effect, practitioners using 0050.TW face a weaker leverage asymmetry, consistent with the insignificant DM test between GJR-GARCH and GARCH reported in Section 4.

3.3 Cross-Market Volatility Spillover

Taiwan’s trading hours (9:00–13:30 local time, or 01:00–05:30 UTC) begin approximately 12 hours after the U.S. market close. This time gap creates a natural information transmission channel. We quantify this spillover using two tests:

Return spillover. The correlation between SPY close-to-close returns on day t and 0050.TW returns on day $t + 1$ is $r = 0.376$, statistically significant ($t = 24.8$, $p < 0.001$). This exceeds the contemporaneous SPY–0050.TW correlation of 0.161, confirming that U.S. information is impounded into Taiwan prices with a one-day lag rather than contemporaneously.

Volatility spillover. Granger causality tests show that lagged VIX significantly predicts 0050.TW squared returns ($F = 58.8$, $p < 0.001$). In contrast, lagged 0050.TW volatility does not Granger-cause VIX ($p = 0.43$), confirming that information flows unidirectionally from the U.S. to Taiwan at the daily frequency. The TWD/USD exchange rate does not add significant explanatory power after controlling for VIX ($p = 0.08$).

These spillover results motivate two distinct strategies: (1) using U.S. VIX as a volatility proxy for Taiwan VT (Section 4), and (2) exploiting the lagged return transmission for time-zone arbitrage (Section 5).

4 Volatility Targeting Strategies for Taiwan

4.1 EWMA Volatility Targeting

The simplest VT approach for Taiwan uses EWMA ($\lambda = 0.94$) to estimate daily volatility and sets the weight according to Equation (4) with a target volatility of 10%. Table 3 reports the results over the out-of-sample period 2010–2026.

EWMA VT improves the Sharpe ratio from 0.729 (buy-and-hold) to 0.796, a modest gain of 0.067. The more dramatic improvement is in maximum drawdown: -41.3% (buy-and-hold) versus -18.4% (EWMA VT), a reduction of 22.9 percentage points. This pattern—modest Sharpe improvement but substantial drawdown reduction—is consistent with findings in the U.S. market [Moreira and Muir, 2017, Harvey et al., 2018] and reflects the mechanical nature of VT: by reducing exposure during high-volatility periods (which tend to coincide with drawdowns), VT truncates the left tail of the return distribution.

A noteworthy finding is that the Sharpe ratio is essentially invariant to the target volatility level: targets from 8% to 18% all produce Sharpe ratios between 0.78 and 0.80. What changes is the risk level: a target of 8% yields a maximum drawdown of -14.2% while a target of 18% yields -28.6% . This invariance implies that investors need not optimize the target—they should simply choose based on their maximum acceptable drawdown.

4.2 GJR-GARCH Versus GARCH

Despite the statistically significant leverage effect in the Taiwan market, GJR-GARCH does not significantly outperform symmetric GARCH in terms of QLIKE forecast accuracy. The Diebold-Mariano test yields $t = -0.17$ ($p = 0.86$), with a QLIKE improvement of only -0.20% . This contrasts sharply with U.S. results, where GJR significantly outperforms GARCH for SPY (DM $t = -6.27$, improvement -3.8% at $w = 2000$).

However, when evaluated in the context of VT portfolio performance, GJR produces a meaningfully higher Sharpe ratio: 1.108 versus 0.994 for GARCH, a difference of +0.114. This disconnect between QLIKE and VT performance arises because VT rewards tail-specific forecast accuracy—the ability to correctly predict when to reduce exposure during sharp declines—rather than average forecast accuracy across all days. GJR’s asymmetric response to negative shocks provides better *directional* timing even when average forecast accuracy is indistinguishable.

This finding has practical implications for model selection: for pure volatility forecasting (e.g., risk reporting), the simpler GARCH model is sufficient for Taiwan. For

portfolio VT, GJR offers a modest but potentially meaningful advantage through improved tail-event response.

Table 3: Volatility Targeting Strategy Performance for 0050.TW

Strategy	Sharpe	MDD (%)	Ann. Return (%)	Ann. Vol (%)	Turnover (%/yr)	Vi
Buy & Hold	0.729	−41.3	10.2	20.8	0	
EWMA VT (10%)	0.796	−18.4	7.8	10.2	116	
GARCH VT (10%)	0.994	−16.8	8.1	10.5	98	
GJR VT (10%)	1.108	−15.1	8.4	10.3	102	
8.63/VIX (monthly)	0.690	−15.3	7.2	12.1	24	
Student- t VaR	—	—	—	—	—	

Notes: Out-of-sample period: 2010–2026 for EWMA VT; 2020–2026 for GARCH/GJR VT (requires 2000-day rolling window from 2008). 8.63/VIX uses monthly rebalancing with 0.585% round-trip transaction cost. VaR violations are the empirical 1% VaR exceedance rate using Student- t (df= 5) distribution over 2020–2026. MDD: maximum drawdown.

4.3 VIX-Proxy Strategy: 8.63/VIX

The VIX-proxy approach sets $w_t = \min(8.63/\text{VIX}_{t-1}, 1)$, where VIX_{t-1} is the U.S. VIX observed at the prior day’s close. This ensures that the weight is determined before the Taiwan market opens, eliminating any contemporaneous timing bias [Moreira and Muir, 2017].

Over 2016–2026, the 8.63/VIX strategy produces a Sharpe ratio of 0.690 and a maximum drawdown of −15.3%, compared to buy-and-hold values of 0.729 and −37.6%. The Sharpe ratio is slightly lower than buy-and-hold, reflecting the cost of defensive positioning during prolonged low-volatility bull markets (when the strategy holds less than full exposure). The maximum drawdown improvement of 22.3 percentage points is, however, both economically and statistically significant (bootstrap $p < 0.001$; see Section 7).

An important methodological consideration is the following. We initially reported Sharpe ratios of 1.16 (daily rebalancing without full transaction costs) and 1.32 (same-

day VIX timing), both of which proved upwardly biased. The daily rebalancing version incurs excessive transaction costs at Taiwan’s 0.585% round-trip rate, and the same-day timing version suffers from a contemporaneous correlation bias documented by [Moreira and Muir \[2017\]](#): since VIX and SPY returns are positively correlated contemporaneously ($\rho \approx 0.65$), using VIX_t to size r_t (rather than r_{t+1}) artificially inflates the Sharpe ratio by approximately 1.0 units. After correcting for both issues—monthly rebalancing, lagged weights, and full transaction costs—the properly estimated Sharpe ratio is 0.690. This reconciliation underscores the importance of strict adherence to lagged-weight protocols in VT backtesting.

Importantly, the Sharpe ratio of the VIX-proxy strategy is independent of the calibration constant K : mathematically, $\text{Sharpe}(K \cdot r) = \text{Sharpe}(r)$ for any positive constant K . The constant $K = 8.63$ affects only the risk level (and hence maximum drawdown), not the risk-adjusted return. This eliminates any concern about calibration leakage from the VIXTWN-to-VIX ratio.

5 Time-Zone Arbitrage

5.1 Motivation and Hypothesis

The cross-market spillover documented in Section 3.3 suggests that U.S. market information is impounded into Taiwan prices with a lag. If this lag is systematic and predictable, it creates a potential trading opportunity. We hypothesize that the sign and magnitude of recent U.S. equity returns predict next-day Taiwan returns, and that this predictability can be exploited through a momentum-based switching strategy.

Specifically, our time-zone arbitrage strategy operates as follows: if the trailing 10-day cumulative SPY return is positive, hold 0050.TW; otherwise, hold cash (or a money market fund). We evaluate lookback windows of 1 to 22 days and report results for the 10-day specification, which balances signal quality against transaction costs: the Taiwan Sharpe ratio improves by 29% (from 1.141 to 1.473) with 38% fewer switches per year (29 versus 47). We acknowledge that the lookback selection introduces a degree of

data mining; however, all eight lookback windows from 1 to 22 days produce positive net Sharpe ratios, and four of eight exceed the Harvey threshold. The 10-day horizon captures the information transmission dynamics more cleanly, as daily noise obscures the underlying signal.

5.2 Main Results

Table 4 presents the time-zone arbitrage results for Taiwan, Japan, and combined portfolios.

Table 4: Time-Zone Arbitrage Strategy Performance

Strategy	Sharpe	t -stat	MDD (%)	Switches/yr	Period	N
<i>Panel A: Individual Markets (10-day SPY Momentum)</i>						
Taiwan (0050.TW)	1.473	3.76	−12.8	29	2012–2025	3,421
Japan (Nikkei 225)	1.306	3.34	−14.5	28	2012–2025	3,267
Hong Kong (HSI)	—	4.12	—	—	2012–2025	—
Australia (ASX 200)	—	4.04	—	—	2012–2025	—
Singapore (STI)	—	4.03	—	—	2012–2025	—
South Korea (KOSPI)	—	3.83	—	—	2012–2025	—
<i>Panel B: Combinations</i>						
TW + JP 50/50	1.810	4.86	−11.9	—	2012–2025	—
Global (US VT + TW TZ)	1.610	4.31	−8.4	—	2012–2025	—
<i>Panel C: Controls</i>						
EWT (US-listed TW ETF)	—	< 1.96	—	—	2012–2025	—
India (Sensex)	—	< 1.96	—	—	2012–2025	—
Indonesia (JCI)	—	< 1.96	—	—	2012–2025	—

Notes: Panel A reports results for locally traded indices/ETFs. t -statistics are computed as $t = \bar{r}_{\text{excess}}/\text{SE}_{\text{NW}}$, where SE_{NW} denotes Newey-West HAC standard errors with automatic bandwidth selection, and compared against the $t > 3.0$ threshold of [Harvey et al. \[2016\]](#). The Newey-West adjustment accounts for serial dependence in strategy returns, yielding lower t -statistics than the naïve Sharpe $\times \sqrt{N_{\text{years}}}$ formula (e.g., for Taiwan 10d: naïve $t = 1.473 \times \sqrt{13} = 5.31$ versus NW-adjusted $t = 3.76$). Panel B reports equal-weighted combinations. Panel C reports control tests: EWT is the iShares MSCI Taiwan ETF traded in U.S. hours (no time-zone gap); India and Indonesia have partial trading overlap with the U.S. Transaction costs of 0.3% per switch (conservative for 0050.TW) are applied throughout.

The Taiwan 10-day SPY momentum strategy generates a net Sharpe ratio of 1.473 after transaction costs (0.3% per switch, approximately 29 switches per year), with a

t -statistic of 3.76, exceeding the [Harvey et al. \[2016\]](#) threshold. The maximum drawdown is -12.8% , a dramatic improvement over buy-and-hold (-41.3%). Japan produces a comparable Sharpe of 1.306 ($t = 3.34$), and four additional Asia-Pacific markets (Hong Kong, Australia, Singapore, South Korea) also exceed the $t > 3.0$ threshold.

5.3 Source of Alpha: Time-Zone Gap, Not Factor Exposure

A critical question is whether the time-zone arbitrage alpha reflects genuine information transmission or merely common factor exposure (e.g., global equity beta). We address this with three tests:

EWT control. The iShares MSCI Taiwan ETF (EWT) trades on the New York Stock Exchange during U.S. hours. If the alpha were driven by common factor exposure, the same momentum signal applied to EWT should produce comparable results. It does not: the EWT control fails to achieve statistical significance ($t < 1.96$). This confirms that the alpha originates from the time-zone information gap—the local market’s delayed incorporation of U.S. information—rather than from mechanical factor exposure.

Europe-to-U.S. test. If time-zone arbitrage is a universal phenomenon, European close-to-U.S.-open signals should also be profitable. We test whether lagged European returns predict next-day U.S. returns and find a negative correlation ($r = -0.07$), the opposite of the predicted sign. This null result is consistent with our mechanism: the U.S. and European markets have substantial trading-hour overlap (approximately 2.5 hours), which largely eliminates the information gap. Time-zone arbitrage requires *zero* trading overlap.

India and Indonesia controls. India (NSE) and Indonesia (JCI) have partial overlap with U.S. futures trading or limited after-hours information flow. Both fail to exceed the $t > 3.0$ threshold, further supporting the zero-overlap requirement.

5.4 Robustness

We subject the time-zone arbitrage result to five robustness tests.

Multi-period out-of-sample. We divide the sample into five non-overlapping sub-

periods (approximately three years each). The strategy produces positive Sharpe ratios in all five periods. The combined t -statistic across periods is 4.47–4.60, depending on the aggregation method.

Bootstrap confidence intervals. Using 10,000 bootstrap replications (block bootstrap with block length of 20 days to preserve serial dependence), the 95% confidence interval for the Sharpe ratio is [0.65, 2.24], excluding zero.

Bonferroni correction. With six markets tested, the Bonferroni-adjusted significance threshold is $t > 3.3$ (assuming $\alpha = 0.05$ and six independent tests). Taiwan ($t = 3.76$) and Japan ($t = 3.34$ for the 10-day lookback; $t = 3.69$ for the 5-day lookback), as well as Hong Kong, Australia, Singapore, and South Korea, all pass this adjusted threshold.

Transaction cost sensitivity. At the observed 29 switches per year and a conservative 0.3% per switch, annual transaction costs are approximately 1.7%. The breakeven cost is over 3% per switch, providing a comfortable margin.

Lookback sensitivity. The 5-day lookback produces a Taiwan Sharpe of 1.141 (NW-adjusted $t = 3.25$); the 10-day lookback produces 1.473 (NW-adjusted $t = 3.76$). Both exceed the [Harvey et al. \[2016\]](#) threshold. The 20-day lookback deteriorates, suggesting that the optimal information horizon is 1–2 trading weeks.

5.5 Combination Strategies

The low correlation between the Taiwan and Japan time-zone signals ($\rho = 0.484$) and between the Taiwan time-zone signal and U.S. VT returns ($\rho = 0.14$) suggests substantial diversification benefits from combining strategies.

TW + JP 50/50. An equal-weighted combination of Taiwan and Japan time-zone arbitrage produces a Sharpe ratio of 1.81 ($t = 4.86$), the highest among all strategies considered, with a maximum drawdown of -11.9% .

Global composite (US VT + TW TZ). Combining a U.S. 12/VIX VT strategy with the Taiwan time-zone strategy produces a Sharpe ratio of 1.61 ($t = 4.31$) and a maximum drawdown of only -8.4% —the lowest drawdown among all strategies.

These combination results exploit the near-orthogonality of the two alpha sources: U.S. VT is driven by variance management (reducing exposure during high VIX), while Taiwan TZ is driven by cross-market momentum (following U.S. trends). The two signals rarely conflict, and when they do (e.g., a VIX spike accompanied by positive SPY momentum), the conservative VIX signal dominates through position sizing.

6 Macroeconomic Indicators for Taiwan Volatility

While our three core contributions focus on leverage amplification, VT strategies, and time-zone arbitrage, we briefly examine whether Taiwan-specific macroeconomic indicators provide incremental volatility forecasting power beyond VIX. An important question for practitioners is whether Taiwan-specific macroeconomic variables can enhance volatility forecasts or VT strategies beyond what VIX alone provides. We conduct a comprehensive sweep of 27 Taiwanese macroeconomic indicators within a GARCH-MIDAS framework, testing each variable’s incremental contribution to 0050.TW volatility forecasting.

6.1 Import Growth as a Volatility Predictor

Among the 27 indicators tested, only one achieves out-of-sample significance: year-over-year import growth (partial $r = 0.214$, OOS improvement +5.6% over base GJR-GARCH, Diebold-Mariano $p = 0.043$). Taiwan is a highly trade-dependent economy, with imports closely tracking global semiconductor demand and industrial production. When import growth exceeds 20%—indicating overheated external demand—adding a “macro guard” that reduces VT exposure improves the Sharpe ratio by 0.056 but worsens maximum drawdown by 3.2 percentage points, an unfavorable tradeoff.

6.2 Null Results

The following indicators fail to improve out-of-sample volatility forecasts or VT performance after controlling for VIX:

- **Business cycle indicator levels:** The National Development Council’s composite business cycle score (“traffic light” system) has no predictive power for next-month realized volatility ($t = -0.53$, $p = 0.60$), consistent with its well-documented lagging nature. Industrial production and M2 money supply growth likewise show insignificant incremental explanatory power (partial $r < 0.05$). However, the *directional change* in these indicators tells a different story; see Section 6.3 below.
- **Margin trading:** Net margin balance and short interest do not predict future volatility or returns. Margin trading in Taiwan appears to be a coincident or lagging indicator rather than a leading indicator.
- **Foreign institutional flows:** Net foreign buying/selling of Taiwanese equities is a lagging indicator, reacting to price movements rather than preceding them.
- **Put/call ratio:** The Taiwan options market put/call ratio shows a contemporaneous correlation of $r = 0.41$ with same-day volatility, but the lagged version drops to zero, confirming that it contains no predictive information.
- **TSMC valuation:** As the dominant constituent of 0050.TW, TSMC’s price-earnings ratio might be expected to predict volatility. It does not: the relationship is driven by the AI-related structural break in TSMC’s earnings, which invalidated historical PE-volatility patterns.

These null results are consistent with our broader finding that VIX serves as a “sufficient statistic” for volatility-related information in markets with strong U.S. linkages. For U.S. equities, we have documented that VIX absorbs the predictive content of more than ten alternative indicators (credit spreads, yield curves, VVIX, MOVE, CNN Fear & Greed Index, AAI sentiment, VIX term structure). For Taiwan, VIX is not perfectly sufficient—import growth and business cycle momentum provide incremental value—but the gap is small enough that the practical benefit of monitoring additional indicators is modest for most investors.

6.3 Business Cycle Indicator Momentum

Beyond trade data, we examine Taiwan’s composite business cycle indicators published by the National Development Council. As noted above, the business cycle score itself has no predictive power for next-month realized volatility ($t = -0.53$, $p = 0.60$), consistent with its well-documented lagging nature. However, the month-over-month change in the leading indicator composite achieves the strongest predictive signal among all Taiwan-specific variables ($t = 3.74$, $p < 0.001$, $R^2 = 7.1\%$), surpassing even trade data. A coincident indicator momentum strategy—reducing equity exposure when the coincident composite declines for three consecutive months—yields Sharpe 0.732 (OOS 2018–2024: 1.260), substantially exceeding the 8.63/VIX baseline of 0.334 over the same 2018–2024 sub-period (compared to 0.690 over the full 2016–2026 sample; see Table 5).¹ The key insight is that the business cycle indicator’s value lies in its *directional change*, not its level.

This finding has broader methodological implications: macroeconomic indicators that appear uninformative in level form may contain substantial predictive content in their first differences. The leading indicator momentum captures regime shifts in the Taiwanese economy—transitions between expansion and contraction—that are orthogonal to the high-frequency volatility information embedded in VIX.

7 VaR and Risk Management

7.1 VaR Methodology

We compute one-day Value-at-Risk at the 1% level using GJR-GARCH conditional volatility combined with the Student- t distribution (degrees of freedom = 5):

$$\text{VaR}_{1\%,t} = \hat{\sigma}_t \cdot q_{0.01}^{t(5)} \quad (7)$$

¹The difference between sub-period and full-sample Sharpe ratios for the 8.63/VIX strategy reflects the strategy’s underperformance during the prolonged low-volatility bull market of 2020–2021, when the VIX-scaled weight held substantially less than full exposure.

where $q_{0.01}^{t(5)} = -3.365$ is the 1% quantile of the standardized Student- t distribution with 5 degrees of freedom.

We also evaluate the skewed Student- t distribution, which simultaneously estimates degrees of freedom (η) and skewness (λ) via maximum likelihood, adapting to each asset's specific tail behavior. For 0050.TW, the estimated parameters are $\eta = 5.2$ and $\lambda = -0.05$ (near-symmetric with moderate fat tails).

7.2 Backtest Results

Over the out-of-sample period 2020–2026 (1,501 trading days), the GJR-GARCH + Student- $t(5)$ VaR produces 8 violations (0.5%), within the Basel III Green Zone threshold of 1.0% (noting that the traffic-light framework is defined for 250-day windows; we apply the 1% threshold as a benchmark across our full sample). The Kupiec likelihood ratio test does not reject correct unconditional coverage ($p = 0.12$). For comparison, the Normal distribution produces 30 violations (2.0%), a borderline result that would trigger the Yellow Zone under Basel III.

The EWMA + Student- $t(5)$ combination performs comparably, with 8 violations (0.5%), demonstrating that the simpler EWMA model is sufficient for regulatory VaR compliance in the Taiwanese market. This is a practical finding: Taiwan investors can implement a fully compliant VaR framework using only EWMA (no GARCH optimization required) plus the Student- t critical value.

7.3 Skewed Student- t for Taiwan

The skewed Student- t distribution passes the Kupiec test for all six assets in our broader cross-asset panel (SPY, QQQ, GLD, TLT, EEM, and 0050.TW)—the only distribution to achieve a perfect 6/6 pass rate. The Cornish-Fisher expansion achieves 5/6 (failing for QQQ due to excessive conservatism), while the fixed Student- $t(5)$ achieves 4/6 and the Normal distribution achieves only 1/6. For Taiwan specifically, both skewed Student- t and fixed Student- $t(5)$ perform well, reflecting the near-symmetric conditional return distribution of 0050.TW.

7.4 Cornish-Fisher Divergence

An important caveat for the Taiwanese market is that the Cornish-Fisher (CF) VaR expansion diverges when using a rolling window of $w = 2000$ for 0050.TW. The issue is that kurtosis estimates become extremely large (up to 545 in some windows) due to occasional extreme returns that are not washed out by the long window. With $w = 504$, CF-VaR performs well ($p = 0.88$), but the shorter window introduces persistence bias in the GARCH estimates. Winsorization of extreme kurtosis values resolves the divergence, but this adds an ad hoc element that undermines the method’s theoretical appeal. For practical purposes, we recommend the skewed Student- t approach for Taiwan VaR, which is both theoretically principled and computationally stable.

8 Discussion

8.1 Currency Risk

An important practical consideration for Taiwanese investors considering U.S.-linked strategies is currency risk. The TWD/USD exchange rate introduces additional volatility that reduces the Sharpe ratio of SPY-denominated strategies by approximately 18% when converted to TWD terms. However, the USD tends to appreciate during equity market crises (a safe-haven effect), providing a partial natural hedge: during the 2022 drawdown, USD appreciation added approximately 15% to returns measured in TWD.

The 8.63/VIX strategy on 0050.TW (Sharpe 0.69 in TWD, no currency risk) modestly outperforms the 12/VIX strategy on SPY converted to TWD (Sharpe 0.67 including FX effects). Given the additional complexity and cost of currency hedging (estimated at approximately 2% per annum for TWD/USD forwards), we do not recommend FX hedging for Taiwanese investors implementing VIX-based strategies on local equities.

8.2 ETFs Versus Individual Stocks for Momentum Strategies

The time-zone arbitrage strategy performs more consistently on the 0050.TW ETF than on individual Taiwanese stocks. Of the ten individual stocks tested, four pass the [Harvey et al. \[2016\]](#) $t > 3.0$ threshold: Yuanta High Dividend ETF (0056.TW, $t = 3.44$), Hon Hai (2317.TW, $t = 3.40$), 0050.TW ($t = 3.25$), and Cathay Financial (2882.TW, $t = 3.06$). However, individual stock results exhibit higher variance and are more susceptible to idiosyncratic events (e.g., Chunghwa Telecom shows a *negative* response to U.S. momentum, $t = -1.17$, reflecting its status as a defensive, dividend-oriented stock that benefits from risk-off flows).

ETFs offer a more stable substrate for the momentum strategy because: (1) diversification smooths idiosyncratic noise, (2) the ETF's larger float reduces market impact costs, and (3) the ETF mechanically reflects the broad market's response to U.S. information, which is the channel the strategy exploits. We therefore recommend 0050.TW (or 0056.TW for income-oriented investors) as the preferred implementation vehicle.

8.3 Practical Implications

Our results suggest several practical implications for individual investors in Taiwan, ordered by implementation complexity:

VIX Step Rule. The simplest approach requires no computation: if $VIX < 15$, allocate 100% to 0050.TW; if $VIX \in [15, 25]$, allocate 70%; if $VIX > 25$, allocate 40%. This captures the bulk of VT's drawdown protection with monthly rebalancing.

8.63/VIX strategy. Compute $w = 8.63/VIX$ at each month-end, allocating $w\%$ to 0050.TW and $(1 - w)\%$ to a short-term bond fund. Transaction cost: approximately 0.15% per rebalancing.

Time-zone momentum. Check whether the trailing 10-day SPY cumulative return is positive. If yes, hold 0050.TW; if no, hold cash. This requires daily monitoring and generates approximately 29 switches per year (transaction cost approximately 1.7% per year).

Combined strategy. Overlaying VIX-based position sizing with momentum switch-

ing produces the lowest drawdown (-8.4%) but requires monitoring two markets and managing two positions.

8.4 Comparison to Prior Literature

Our findings both confirm and extend the existing VT literature. Consistent with [Moreira and Muir \[2017\]](#), VT’s primary benefit is drawdown reduction rather than Sharpe ratio improvement; the Sharpe improvement t -statistic is only 0.33, far below conventional significance levels, but the MDD improvement is statistically significant (bootstrap $p = 0.0004$). This result holds in both U.S. and Taiwanese markets, suggesting it is a universal property of VT strategies.

Our time-zone arbitrage finding relates to the broader literature on cross-market momentum and information transmission. [Rapach et al. \[2013\]](#) document that U.S. stock returns predict returns in non-U.S. markets, attributing this to the U.S. market’s role as a global information leader. Our contribution is to show that this predictability is economically significant for *locally traded* Asian equities (but not for U.S.-listed ADRs/ETFs of the same underlying markets), and that the zero trading-hour overlap condition is necessary for the effect to be exploitable. The failure of the European-to-U.S. channel, where trading hours overlap, provides a natural placebo test.

The diversification amplification of the leverage effect extends the work of [Ang and Chen \[2002\]](#) on asymmetric correlations. While Ang and Chen document that correlations increase during downturns, we show that this phenomenon mechanically amplifies the GJR-GARCH γ parameter at the index level relative to the constituent level, and that the amplification ratio varies across markets ($4.6\times$ for the broad TAIEX, $1.45\times$ for the investable 0050.TW, versus $2.8\times$ for the U.S.). The gap between TAIEX and 0050.TW ratios highlights the importance of distinguishing between broad market indices and investable vehicles when quantifying the leverage effect.

8.5 Combining Macro Signals with VIX-Based Strategies

A natural question is whether Taiwan-specific macro signals can enhance the VIX-based framework. We find that combining VIX scaling with the leading indicator direction—using $k = 10$ (aggressive) when the leading indicator is rising and $k = 6$ (conservative) when declining—significantly improves the pure 8.63/VIX strategy over the full 2012–2026 sample (Diebold-Mariano $p = 0.0005$, Sharpe from 0.999 to 1.151; see Table 5). The intuition is straightforward: when the leading indicator signals economic expansion, the VT strategy can afford to take more risk; when contraction looms, it should be more defensive. This conditional approach effectively introduces regime awareness into an otherwise purely volatility-driven framework.

However, this monthly-frequency macro signal cannot improve the daily-frequency SPY momentum strategy (DM $p = 0.82$), confirming that effective signal combination requires matching timescales. The SPY momentum signal already captures high-frequency regime information through daily U.S. market movements, leaving no room for a slower-moving macro indicator to add value. This timescale-matching constraint has practical implications: investors using the time-zone arbitrage strategy should not attempt to overlay macro filters, while investors using the monthly VIX-rebalancing strategy can benefit meaningfully from incorporating leading indicator direction.

8.6 Sharpe Ratio Reconciliation

Table 5 reconciles the Sharpe ratios reported throughout the paper for the 8.63/VIX strategy and its variants. The apparent discrepancies arise from differences in sample period, rebalancing frequency, and strategy specification (pure VIX scaling versus the VIX + leading indicator combination).

Table 5: Sharpe Ratio Reconciliation for VIX-Based Strategies on 0050.TW

Strategy	Period	Sharpe	Table/Section
8.63/VIX (monthly, lagged)	2016–2026 (full)	0.690	Table 3
8.63/VIX (monthly, lagged)	2018–2024 (sub-period)	0.334	Section 6.3
8.63/VIX + Leading indicator	2012–2026 (full)	0.999	Section 8
VIX + Leading indicator combo	2012–2026 (full)	1.151	Section 8

Notes: All strategies use monthly rebalancing with lagged weights (VIX_{t-1} determines the weight for period t) and include transaction costs of 0.585% round-trip. The 2018–2024 sub-period Sharpe is lower because this window captures the prolonged low-volatility bull market of 2020–2021, during which the VIX-scaled strategy was underinvested relative to buy-and-hold. The VIX + Leading indicator combination uses $k = 10$ when the leading indicator rises and $k = 6$ when it declines; its longer sample (2012–2026) reflects the availability of leading indicator data from 2012.

9 Conclusion

This paper examines volatility targeting strategies in the Taiwan stock market, contributing three sets of findings.

First, we document that the TAIEX exhibits a strong leverage effect ($\gamma = 0.272$, $t = 3.18$) that is amplified $4.6\times$ relative to individual stock asymmetries for the broad index (~ 900 stocks), though the investable 0050.TW (50 stocks, $\gamma = 0.087$) shows a more modest $1.45\times$ amplification. Both exceed the U.S. amplification ratio of $2.8\times$ at the broad-index level. This amplification arises from higher correlation asymmetry among Taiwanese equities during market declines and has direct implications for model selection: index-level strategies should account for asymmetric volatility even when individual constituents show weak asymmetry.

Second, we evaluate EWMA, GARCH, GJR-GARCH, and VIX-proxy VT strategies adapted for the Taiwan market. EWMA VT improves the Sharpe ratio from 0.729 to 0.796 while halving maximum drawdown from -41.3% to -18.4% . The VIX-proxy strat-

egy (8.63/VIX), calibrated using the VIXTWN-to-VIX ratio of 1.39, achieves comparable drawdown protection with only monthly rebalancing. A key finding is the disconnect between forecast accuracy and portfolio performance: GJR-GARCH does not significantly outperform GARCH in QLIKE (DM $p = 0.86$), yet GJR-based VT produces a meaningfully higher Sharpe ratio (+0.114), suggesting that tail-specific accuracy matters more than average accuracy for VT applications.

Third, we document and evaluate a time-zone arbitrage strategy: a 10-day SPY momentum signal applied to 0050.TW generates a net Sharpe ratio of 1.473 with a [Harvey et al. \[2016\]](#) t -statistic of 3.76. This effect extends to six Asia-Pacific markets (all exceeding $t > 3.0$) and survives multi-period out-of-sample tests, bootstrap confidence intervals ([0.65, 2.24]), and Bonferroni correction. The failure of U.S.-listed ETF controls and the European-to-U.S. channel confirms that zero trading-hour overlap is a necessary condition. A combined U.S. VT and Taiwan time-zone strategy achieves a Sharpe ratio of 1.61 with a maximum drawdown of only -8.4% .

Several limitations warrant mention. First, the VIXTWN data history is limited to 64 months (November 2020 to present), which restricts our ability to validate the VIX-to-VIXTWN ratio across different market regimes. As VIXTWN data accumulates, the calibration should be periodically updated. Second, the time-zone arbitrage strategy requires daily monitoring and generates approximately 29 switches per year, which may not suit all retail investors. Third, our VaR analysis uses a rolling window of 2,000 days, which may be slow to adapt to structural breaks in the Taiwanese market; shorter windows (504 days) improve VaR compliance but introduce GARCH persistence bias. Fourth, we test only a single emerging Asian market in depth; replication in other markets with similar characteristics (e.g., South Korea, where preliminary results are promising) would strengthen the generalizability of our findings.

Future research directions include: (1) validating the VIXTWN-based strategy when sufficient data become available (currently 64 months; 252+ months recommended for comprehensive evaluation); (2) extending the time-zone arbitrage analysis to intraday data, which could reveal whether the information gap closes within the first minutes

of Taiwan trading; and (3) investigating whether the diversification amplification phenomenon has increased over time as passive ETF investing has grown in Taiwan, potentially strengthening the leverage effect at the index level.

References

- Ang, A. & Chen, J. (2002). Asymmetric correlations of equity portfolios. *Journal of Financial Economics*, 63(3), 443–494.
- Black, F. (1976). Studies of stock price volatility changes. *Proceedings of the 1976 Meetings of the American Statistical Association, Business and Economic Statistics Section*, 177–181.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.
- Bozovic, M. (2024). VIX-managed portfolios. *International Review of Financial Analysis*, 95, 103353.
- Christie, A.A. (1982). The stochastic behavior of common stock variances: Value, leverage and interest rate effects. *Journal of Financial Economics*, 10(4), 407–432.
- Diebold, F.X. & Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- Fleming, J., Kirby, C., & Ostdiek, B. (2001). The economic value of volatility timing. *Journal of Finance*, 56(1), 329–352.
- Glosten, L.R., Jagannathan, R., & Runkle, D.E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *Journal of Finance*, 48(5), 1779–1801.
- Harvey, C.R., Liu, Y., & Zhu, H. (2016). ...and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.

- Harvey, C.R., Hoyle, E., Korgaonkar, R., Rattray, S., Sargaison, M., & Van Hemert, O. (2018). The impact of volatility targeting. *Journal of Portfolio Management*, 45(1), 14–33.
- Hwang, S. & Valls Pereira, P.L. (2006). Small sample properties of GARCH estimates and persistence. *European Journal of Finance*, 12(6–7), 473–494.
- Moreira, A. & Muir, T. (2017). Volatility-managed portfolios. *Journal of Finance*, 72(4), 1611–1644.
- Patton, A.J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256.
- Rapach, D.E., Strauss, J.K., & Zhou, G. (2013). International stock return predictability: What is the role of the United States? *Journal of Finance*, 68(4), 1633–1662.